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The Quantum Trace of Frobenius as a Formal Object

Definitions, proofs and reproducible validation

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Abstract

We distinguish an intrinsic normalized Frobenius power trace, a controlled-circuit access contract, and a quantum estimator. For a smooth projective curve, the quantity exists unconditionally and is the trace of a unitary symplectic realization of normalized Frobenius. A phase-tolerant contract supplies the controlled operators needed for coherent amplitude estimation, including an explicit error budget for approximate implementations. We define a quantum trace estimate by its statistical guarantee and prove reconstruction below a congruence threshold. Sharpness is established for bounded-error records of abstract Weil polynomials through order $2g$, with separately checked examples realized by curves. We prove fixed-point and route-specific precision statements for the point-permutation unitary. With a stated accuracy margin, computable gates, and logarithmically many ancillas, a classical circuit description allows classical recovery of the Frobenius polynomial. The resulting general classical hardness conjecture remains open. Seven Python scripts check finite instances, including regressions for previously overstated claims; they do not prove universal statements or complexity conjectures.

Validation code. Every result in this white paper can be checked with Python validation scripts, which are published on GitHub at:
https://github.com/teknoquants/Quantum-Trace-Estimates-for-Frobenius-Mathematical-Foundations-Circuit-Access-and-Validation

Main distinction. The number

$$\tau_j(C) = \frac{Q^j + 1 - \#C(\mathbb{F}_{Q^j})}{Q^{j/2}}$$

is an arithmetic invariant. The word *quantum* describes an estimation procedure with specified access and uncertainty. It does not add a new arithmetic invariant, nor does mathematical existence of a unitary establish an efficient circuit construction.

Status and attribution. The Weil theorem, semisimplicity of Frobenius on the Jacobian, and amplitude estimation are standard inputs, cited below. This paper gives explicit deductions and a formal organization, without a claim of priority for those inputs. The supplied manuscripts [8, 9, 10] motivate the notation and congruence decoder; all deductions needed here are proved in this edition. Definitions stipulate objects; their consequences are the propositions and theorems. The observation about the point-permutation route is attributed to Roger Cooperman; this is an attribution of the observation, not a verified historical priority claim.

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1 Conventions and scope

Let $C/\mathbb{F}_Q$ be smooth, projective and geometrically connected of genus $g \geq 1$, where $Q = p^r$. Put $N_j = \#C(\mathbb{F}_{Q^j})$, including points at infinity, and

$$Z_C(T) = \exp\left(\sum_{j\geq 1} N_j T^j/j\right) = \frac{L_C(T)}{(1-T)(1-QT)}, \quad L_C(T) = \sum_{k=0}^{2g} b_k T^k = \prod_{i=1}^{2g}(1-\alpha_i T).$$

The positive real square root is used throughout. The Weil theorem gives $b_k \in \mathbb{Z}$, $b_0 = 1$, $|\alpha_i| = \sqrt{Q}$ in every complex embedding, and

$$b_{2g-k} = Q^{g-k} b_k \quad (0 \leq k \leq g). \tag{1}$$

We use geometric Frobenius F on $H^1_{\text{ét}}(C_{\overline{\mathbb{F}}_Q}, \mathbb{Q}_\ell)$, $\ell \neq p$, with $\det(1 - TF) = L_C(T)$. Here and below $\mathbb{Q}_\ell$ denotes the ℓ-adic rational field. An abstract embedding into $\mathbb{C}$ is used only to discuss a complex realization, not as a computational procedure.

The class $W_g(Q)$ consists of integer polynomials of degree $2g$ satisfying (1), $b_0 = 1$, and the Weil root condition. It is larger than the set proved to arise from curves; no assertion of universal curve realizability is made.

The complexity claims involving input curves use an explicit representation whose length is polynomial in g and $\log Q$. In particular, Conjecture 1 concerns dense squarefree hyperelliptic equations over odd-characteristic fields, with a supplied polynomial-basis field representation. Arithmetic and gate descriptions are counted in bits. Genus zero has $L_C = 1$ and $\tau_j = 0$ and needs no quantum estimator.

2 Step 1 The intrinsic quantity

Definition 1 (Frobenius power trace)**.** For $j \geq 1$, define $s_j(C) = Q^j + 1 - N_j$ and $\tau_j(C) = Q^{-j/2} s_j(C)$.

Proposition 1 (A quantity defined by the curve alone)**.** *For all $j \geq 1$,*

$$s_j(C) = \mathrm{Tr}(F^j \mid H^1) = \sum_i \alpha_i^j, \qquad \tau_j(C) = \sum_i (\alpha_i/\sqrt{Q})^j \in \mathbb{R}, \qquad |\tau_j(C)| \leq 2g.$$

The integer s_j and real number τ_j are invariant under $\mathbb{F}_Q$-isomorphism and independent of ℓ and the complex embedding. The first g traces determine L_C and all later traces.

Proof. The trace formula, or equivalently the logarithm of the zeta-function factorization, gives $N_j = 1 + Q^j - \sum_i \alpha_i^j$; see [1]. Integrality follows from the definition of s_j, and reality follows from complex conjugation of the roots of $L_C \in \mathbb{Z}[T]$. The Weil bound follows from $|\alpha_i/\sqrt{Q}| = 1$. An isomorphism preserves every extension-field point count. The integer polynomial L_C fixes the complex multiset of roots independently of the auxiliary realization. Finally Newton's identities give

$$k b_k = -s_k - \sum_{i=1}^{k-1} s_i b_{k-i}, \qquad b_0 = 1. \tag{2}$$

These determine $b_1, \ldots, b_g$, and (1) supplies the rest. Newton's recurrence then gives every later s_j. □

For a quadratic twist $y^2 = df(x)$ with d a nonsquare in $\mathbb{F}_Q$, s_j changes by $(-1)^j$. Indeed d remains nonsquare in $\mathbb{F}_{Q^j}$ exactly for odd j; the quadratic-character count, including the infinity

contribution, gives this identity. In the scripts, coefficients are in the prime field; the $Q = 9$ examples are base changes, not an enumeration of all curves with coefficients in $\mathbb{F}_9$.

Validation v1. Exhaustive finite-field counts, exact Newton reconstruction, exact Weil membership, isomorphic coordinate changes, and twists are checked. Numerical spectral comparisons are tolerance-based checks, not exact algebraic equalities established by floating-point arithmetic.

3 Step 2 A unitary realization and its access model

Extend H^1 to a complex vector space V_C. The cup-product form ω is alternating and nondegenerate and satisfies $\omega(Fx, Fy) = Q\omega(x, y)$. Let $J = \left(\begin{smallmatrix} 0 & I_g \\ -I_g & 0 \end{smallmatrix}\right)$ and $\mathrm{USp}(2g) = U(2g) \cap \mathrm{Sp}_{2g}(\mathbb{C})$.

Lemma 1 (Unitary symplectic realization)**.** *There is a symplectic basis and a Hermitian inner product in which $U_C = Q^{-1/2}F$ is*

$$\mathrm{diag}(\beta_1, \ldots, \beta_g, \overline{\beta}_1, \ldots, \overline{\beta}_g) \in \mathrm{USp}(2g), \quad |\beta_i| = 1.$$

Its jth power has trace $\tau_j(C)$. The set

$$\mathcal{R}(C) = \{U \in U(2g) : \det(I - tU) = L_C(t/\sqrt{Q})\}$$

is a single unitary conjugacy class. Its intersection with $\mathrm{USp}(2g)$ *is a single symplectic-unitary conjugacy class. In particular, the eigenvalues* 1 *and* -1 *have even multiplicity.*

Proof. Frobenius is semisimple on H^1, via the Jacobian and the semisimplicity of its Frobenius endomorphism [2]. Thus U_C is diagonalizable, has eigenvalues on the unit circle, and preserves ω. For eigenvectors in eigenspaces E_β, E_γ, invariance gives $\omega(x, y) = \beta\gamma\omega(x, y)$. The pairing vanishes unless $\gamma = \beta^{-1} = \overline{\beta}$.

For a nonreal pair of eigenspaces, take any basis of E_β and its ω-dual basis in $E_{\overline{\beta}}$. On E_1 and E_{-1} the restricted alternating form is nondegenerate, so these spaces have even dimension and admit symplectic bases. Order all first basis vectors before all their partners. This gives the asserted diagonal matrix and J. Declare the resulting basis orthonormal. The diagonal matrix is unitary and preserves J.

The trace claim is Proposition 1. Two unitary matrices with the same eigenvalue multiset are unitarily conjugate. For the compact symplectic group, diagonal torus elements with the same spectrum differ only by permutation of the pairs and exchanges of partners; these operations have symplectic-unitary representatives. Equivalently this is the maximal-torus conjugacy theorem [3]. The construction supplies a representative, not a preferred basis or a canonical matrix. □

For a unitary X, write $\Lambda_k(X)$ for its k-fold controlled operator. Crucially, $\Lambda_1(e^{i\gamma}X) \neq e^{i\gamma}\Lambda_1(X)$ in general: the former introduces a relative phase.

Definition 2 (Frobenius oracle and access model)**.** Choose $n = 2^q \geq 2g$ and $V = \widetilde{U} \oplus I_{n-2g}$, where $\widetilde{U} \in \mathcal{R}(C)$. An ε-accurate oracle supplies two unitary circuits O_1, O_2, with a private ancilla qubits each, satisfying, for some fixed phases γ_1, γ_2,

$$\|O_k(|\psi\rangle|0^a\rangle) - e^{i\gamma_k}(\Lambda_k(V)|\psi\rangle)|0^a\rangle\| \leq \varepsilon\|\psi\|, \quad k = 1, 2. \tag{3}$$

The inverse circuits are supplied by reversal. An algorithm calls these circuits or their inverses without adding controls to them, and does not operate on their private ancillas between calls. Its ideal execution keeps those ancillas clean. Calls and their types follow a fixed coherent schedule within each estimation run. Known gates and measurements are available separately. Public parameters include g, Q, n and the requested precision and confidence.

The quantifier over every input in (3) is an operator-norm bound on the clean-input subspace, so it also holds with arbitrary reference registers. Exact access means $\varepsilon = 0$. Efficient construction means a uniform algorithm producing circuits with bit length and gate count polynomial in the input length and $\log(1/\varepsilon)$. Mathematical existence does not imply this property. Minimal padding $n = 2^{\lceil \log_2(2g) \rceil} < 4g$ will be stated whenever n is replaced by $O(g)$.

Validation v2. Numerical constructions test the symplectic basis and conjugacy statements. Complete finite Weil-class enumeration and real-root parity use exact integer arithmetic and factorization. The access contract itself is stipulated; v3 tests its consequences, including independently perturbed O_1, O_2 and ancilla leakage.

4 Step 3 The estimator

Prepare $|\Phi_n\rangle = n^{-1/2} \sum_x |x\rangle|x\rangle$. Let A_j prepare $|0\rangle|\Phi_n\rangle$, apply a Hadamard to the control, apply $\Lambda_1(I \otimes V^j)$, and apply a final Hadamard. Acceptance is control outcome zero.

Theorem 1 (Estimator under exact access)**.** *Suppose Definition 2 holds with $\varepsilon = 0$. For $j \geq 1$, $0 < \delta \leq 2n$ and $0 < \eta < 1$:*

(i) The acceptance probability is

$$a_j = \frac{1}{2} + \frac{\tau_j(C) + n - 2g}{2n}. \tag{4}$$

The analogous imaginary-part test accepts with probability $1/2$.

(ii) With $M = 2^{\lceil \log_2(4\pi n/\delta) \rceil}$, canonical amplitude estimation returns $\widetilde{a}$ for which

$$\widehat{\tau}_j = 2n(\widetilde{a} - 1/2) - (n - 2g)$$

is within δ of τ_j with probability at least $p_0 = 8/\pi^2$.

(iii) Take an odd integer $r \geq \log(1/\eta)/(2(p_0 - 1/2)^2)$. The median of r independent estimates has failure probability at most η.

(iv) The stated circuit uses

$$rj(2M - 1) < 2rjM < 16\pi rjn/\delta$$

oracle calls. Its cost is $O(jn\delta^{-1} \log(2/\eta))$, and becomes $O(jg\delta^{-1} \log(2/\eta))$ only for minimal padding (or more generally $n = O(g)$).

Proof. The identity $\langle \Phi_n | (I \otimes X) | \Phi_n \rangle = \mathrm{Tr}(X)/n$ gives the Hadamard acceptance probability. Since $\mathrm{Tr}(V^j) = \tau_j + n - 2g$ is real, (4) and the imaginary-part statement follow.

The amplitude-estimation theorem of Brassard, Høyer, Mosca and Tapp [4] gives, with probability at least $8/\pi^2$,

$$|\widetilde{a} - a| \leq \frac{2\pi\sqrt{a(1-a)}}{M} + \frac{\pi^2}{M^2} \leq \frac{\pi}{M} + \frac{\pi^2}{M^2} \leq \frac{\delta}{2n}.$$

The last inequality follows from $M \geq 4\pi n/\delta$ and $\delta \leq 2n$. Multiplying by $2n$ proves (ii). Hoeffding's inequality bounds the probability that at least half of the independent runs fail by $\exp[-2r(p_0 - 1/2)^2]$, proving (iii).

The preparation A_j uses j calls to O_1. Phase estimation of the Grover iterate uses $M - 1$ controlled iterates. Each contains controlled A_j and controlled $A_j^\dagger$, implemented using j calls each to O_2 and $O_2^\dagger$, together with controlled known gates. Whole-call phases cancel between these paired calls; the preparation phase is global. Thus no extra control on an unknown phased oracle is needed. The count is $j + 2j(M - 1)$ per run. Since $M < 8\pi n/\delta$, the cost bound follows. □

The cost counts oracle invocations, not elementary gates, state preparation, precision in known rotations, or classical postprocessing. Independent Hadamard sampling alone has a sufficient sample count of order n^2/δ^2; the linear inverse-precision cost here uses coherent amplitude estimation.

4.1 Approximate access

Lemma 2 (Error accumulation)**.** *The inverses in Definition 2 satisfy the same clean-input error bound, with phases $-\gamma_k$. An algorithm with a fixed schedule of K oracle calls and ideal-clean private ancillas has final state within $K\varepsilon$ of a global phase times its ideal final state. Any final measurement distribution differs by total variation at most $K\varepsilon$.*

Proof. For the inverse, substitute $\psi = \Lambda_k(V)^\dagger\varphi$ in (3), apply $O_k^\dagger$, and multiply by $e^{-i\gamma_k}$. This gives the bound for $O_k^\dagger$ on every clean input.

For a schedule of gates, telescope the difference of the two products with the ideal prefix on the right of each difference term and the actual suffix on its left. The prefix has clean private ancillas, so each difference has norm at most ε, even with entangled reference registers. The suffix is unitary and preserves the norm. Whole-call phases multiply to one global phase. The triangle inequality gives $K\varepsilon$. Pure-state trace distance, and hence measurement total variation, is at most the distance between the state vectors after optimizing a global phase. □

Theorem 1′ (Estimator with approximate access). Use the M of Theorem 1 and let $K = j(2M-1)$. If $K\varepsilon \le 0.1$, each run succeeds within δ with probability at least $8/\pi^2 - 0.1 > 0.71$. An odd number

$$r \ge \frac{\log(1/\eta)}{2(0.71 - 1/2)^2}$$

of independent runs gives failure probability at most η. A sufficient oracle accuracy is $\varepsilon \le \delta/(160\pi jn)$.

Proof. Lemma 2 reduces success probability by at most $K\varepsilon$. Apply the same median argument and use $K < 16\pi jn/\delta$. □

Validation v3. Hadamard states and amplitude-estimation output laws are checked numerically. Six additional small full-circuit simulations count actual O_1, O_2 and inverse applications and compare their output laws with the analytic formula. Twelve full-circuit cases use independent perturbations of both oracle types, different global phases, and one private ancilla with leakage. Larger Monte Carlo experiments use analytic sampling and formula-based cost bookkeeping, explicitly distinguished from circuit instrumentation. Finite amplitude grids test examples, not a continuum guarantee; the latter follows from the theorem.

5 Step 4 The quantum trace estimate

Definition 3 (Quantum trace estimate)**.** For $j \ge 1$, $\delta > 0$ and $0 < \eta < 1$, a (δ, η) quantum trace estimate of order j is a real random output $\widehat{\tau}_j$ of a quantum algorithm with Frobenius-oracle access and the public parameters of Definition 2, satisfying

$$\Pr\{|\widehat{\tau}_j - \tau_j(C)| \le \delta\} \ge 1 - \eta.$$

A record is a finite list of such estimates together with its orders, error tolerances and failure guarantees. The target is intrinsic; the output distribution, runtime and implementation need not be intrinsic.

Let

$$\delta_{\mathrm{cong}}(g,Q)=\min_{1\le k\le g}\frac{k}{2Q^{k/2}}=\frac{1}{2}\min\{Q^{-1/2},\,gQ^{-g/2}\}. \tag{5}$$

The endpoint identity holds because $\log k-(k/2)\log Q$ is concave in k. In particular, when the second endpoint is smaller the value is exactly $g/(2Q^{g/2})$; a genus-dependent factor must not be suppressed in a uniform asymptotic claim.

Proposition 2 (Well-posedness, existence and reconstruction)**.** *(a) The target and defining error condition do not depend on the auxiliary cohomology realization. Known identity padding is removed by the estimator.*

(b) Conditional on exact access, or on approximate access with the accuracy in Theorem 1′, estimates exist for every requested $0<\delta\le 2g$, $0<\eta<1$.

(c) If orders $1,\dots,g$ are each estimated within $\delta<\delta_{\mathrm{cong}}$ with failure probability at most η, their congruence decoder recovers L_C with probability at least $1-g\eta$. Independence is unnecessary for this conclusion. For a total failure budget η_{tot}, use $\eta=\eta_{\mathrm{tot}}/g$.

(d) For every $g\le m\le 2g$, two distinct polynomials in $W_g(Q)$ have normalized trace records through order m at distance exactly $2\delta_{\mathrm{cong}}$. Consequently a bounded-error record with radius $\delta\ge\delta_{\mathrm{cong}}$ alone need not identify its polynomial. This is a deterministic ambiguity statement about the numerical record guarantee, not a query lower bound or a universal curve-realizability assertion.

Proof. Parts (a) and (b) follow from Proposition 1, Lemma 1 and Theorem 1. For (c), put $\widehat{s}_k=Q^{k/2}\widehat{\tau}_k$. When all estimates succeed, $|\widehat{s}_k-s_k|<k/2$. Suppose the coefficients and power sums through order $k-1$ have been recovered. Write $R_k=\sum_{i<k}s_ib_{k-i}$. Newton's identity forces $s_k\equiv -R_k \pmod k$, so rounding $(-R_k-\widehat{s}_k)/k$ recovers the integer b_k uniquely. Recover $s_k=-R_k-kb_k$ and continue. The functional equation finishes L_C. A union bound gives the stated confidence.

For (d), consider the pairs

$$L_0=\sum_{i=0}^{g}Q^iT^{2i},\qquad L_1=L_0-T\sum_{i=0}^{g-1}Q^iT^{2i},$$
$$M_0=1+Q^gT^{2g},\qquad M_1=M_0+T^g.$$

They satisfy integrality and the functional equation. Put $z=\sqrt{Q}T$ and $c=Q^{-1/2}$. On the unit circle the normalized first pair, divided by z^g, is $U_g(\cos\theta)$ and $U_g(\cos\theta)-cU_{g-1}(\cos\theta)$, where U_k is the Chebyshev polynomial of the second kind. Interlacing of U_g,U_{g-1} places the latter's g roots in $(-1,1)$: the signs alternate at successive roots of U_{g-1}, and the endpoint values at ±1 provide the two outer intervals since $0<c<1$. The $g=1$ case is $2x-c$. Thus both polynomials have all normalized roots on the unit circle. For the second pair, $z^g=y$ reduces the equations to $1+y^2=0$ and $1+Q^{-g/2}y+y^2=0$, whose roots lie on the unit circle. Hence all four belong to $W_g(Q)$.

The first pair satisfies

$$\frac{L_1(z/\sqrt{Q})}{L_0(z/\sqrt{Q})}=1-cz+O(z^{2g+1}).$$

Since normalized traces are the coefficients of $-j[z^j]\log L(z/\sqrt{Q})$, the trace differences are c^j for $1\le j\le 2g$. Their sup-distance is $Q^{-1/2}$. For the second pair the differences through order $2g$ vanish except at g and $2g$, where they are $-gQ^{-g/2}$ and gQ^{-g}, respectively. Its sup-distance is $gQ^{-g/2}$.

Choose the first pair when $Q^{-1/2}\le gQ^{-g/2}$ and the second pair otherwise. It is this choice, not both pairs individually, that attains $2\delta_{\mathrm{cong}}$. Their midpoint is within δ_{cong} of both records, already with zero error probability. A record-only decoder cannot distinguish them. For $m\ge g$

truncation preserves the relevant maximum. This argument concerns the deterministic error condition underlying Definition 3; it does not assert that a member of $W_g(Q)$ is a curve. □

With minimal padding, $\delta = \delta_{\mathrm{cong}}/2$, and individual failure budget η_{tot}/g, the trace route uses at most $rMg(g+1)$ calls. Uniformly in g, Q, its order is

$$O\Big(g^3\delta_{\mathrm{cong}}^{-1}\log(2g/\eta_{\mathrm{tot}})\Big) = O\Big(g^2Q^{g/2}\log(2g/\eta_{\mathrm{tot}})\Big).$$

For the last simplification, $\delta_{\mathrm{cong}}^{-1} = 2\max(\sqrt{Q}, Q^{g/2}/g)$ and $g\sqrt{Q}/Q^{g/2}$ is bounded for $Q \geq 2$, $g \geq 1$. If arbitrary larger padding is allowed, retain n in the cost instead.

5.1 Curve witnesses and the range of orders

Abstract Weil witnesses do not by themselves establish a threshold over every family of curves. Table 1 supplies finite curve-realized examples. Each pair is selected by testing *all* orders $1, \ldots, 2g$, with exact Newton predictions checked against independent exhaustive point enumeration through order $2g$. Squared sup-distances are compared as exact rational numbers. The prior first-g search was insufficient for that claim. The regression test retains the former examples and detects the four failures of their full-order claim.

g	Q	Curve	$f(x)$ in $y^2 = f(x)$	(N_1, N_2)	Distance
1	5	1	$x^3 + x^2 + 2x$	$(6, 36)$	0.447214
1	5	2	$x^3 + x^2 + 1$	$(5, 35)$	
1	7	1	$x^3 + x$	$(8, 64)$	0.377964
1	7	2	$x^3 + 3x^2 + x + 1$	$(9, 63)$	
1	11	1	$x^3 + x$	$(12, 144)$	0.301511
1	11	2	$x^3 + 7x^2 + x + 1$	$(11, 143)$	
1	13	1	$x^3 + x^2 + x$	$(16, 192)$	0.277350
1	13	2	$x^3 + 6x^2 + x + 1$	$(15, 195)$	
2	3	1	$x^6 + x^5 + 2x^4 + x^2 + x$	$(4, 18)$	0.577350
2	3	2	$x^6 + x^4 + x^3 + x^2 + x$	$(5, 17)$	
2	5	1	$x^6 + x^4 + x$	$(5, 25)$	0.400000
2	5	2	$x^6 + 2x^5 + 3x^3 + 2x^2 + x$	$(5, 27)$	

Table 1: Corrected curve pairs. Distance is the sup-distance of normalized traces over all orders $1 \leq j \leq 2g$, equal to $2\delta_{\mathrm{cong}}$. The JSON records exact squared distances and independently enumerated counts through $2g$.

For the abstract pairs, v4 also tests orders through $4g$ and records increased separations beyond $2g$. These examples justify no universal extension of Proposition 2(d) to longer records. No theorem about sharp curve-realized thresholds for all genera or fields is claimed here.

Validation v4. Besides the full-order witness search and regression, the script checks invariance under padding and conjugation of the success guarantee, sampled coverage, congruence decoding, sign-pattern error tests, and midpoint membership in both error balls. A finite sample of successful recovery is evidence for the implementation; the probability guarantee is supplied by the proof.

6 Step 5 The point-permutation unitary

Proposition 3 (Fixed points and the natural permutation)**.** *Let $X_m = C(\mathbb{F}_{Q^m})$, $n_m = |X_m|$, and let P_m act on the point basis of $\ell^2(X_m)$ by $P_m|x\rangle = |\varphi(x)\rangle$, where φ raises coordinates to their*

Qth powers. Then

$$P_m^\dagger P_m = I, \quad P_m^m = I, \quad \mathrm{Tr}(P_m^j) = N_{\gcd(j,m)}.$$

In particular the trace is N_j when $j \mid m$. For each $l \mid m$, the number of orbits of length l is

$$o_l = \frac{1}{l}\sum_{d \mid l} \mu(l/d) N_d.$$

Each such orbit contributes each lth root of unity once to the spectrum.

Proof. The coordinate map fixes coefficients over $\mathbb{F}_Q$, preserves the curve, and its mth iterate is the identity on $\mathbb{F}_{Q^m}$. It is therefore a permutation and its matrix is unitary. A permutation trace is a fixed-point count. A point fixed by φ^j has coordinates in $\mathbb{F}_{Q^m} \cap \mathbb{F}_{Q^j} = \mathbb{F}_{Q^{\gcd(j,m)}}$. This argument is also valid on projective charts. Cycle decomposition gives the spectral assertion and $N_d = \sum_{l \mid d} l o_l$ for $d \mid m$; Möbius inversion gives the formula. □

Coordinate-wise implementation is economical in an explicit finite-field encoding. The map $x \mapsto x^Q$ is an invertible $\mathbb{F}_p$-linear map on $\mathbb{F}_{Q^m}$, of vector dimension rm. Gaussian elimination expresses it in polynomially many reversible modular row operations. On a model with a bounded number of coordinate registers this supplies a polynomial-size ambient coordinate circuit, including the representation's infinity flags. Its restriction to valid point encodings realizes P_m.

This does *not* supply an efficient dense indexing of the n_m points, their count, or a uniform superposition over valid points. Unused computational states in the ambient coordinate space must not be included in the point trace. Thus easy coordinate evolution alone does not implement the entire normalized trace-estimation route. These encoding and state-preparation qualifications are essential.

Validation v5. Explicit permutations, cycle spectra, Möbius orbit counts, and prime-field linear maps are tested, including the swapped infinity points of an even-degree model. The script does not construct an efficient uniform state over all curve points.

6.1 Precision and cost for the specified route

Proposition 4 (The point-permutation trace route)**.** *Assume $n_m = N_m > 0$, $1 \le j < m$, and $j \mid m$. Suppose the route is supplied with n_m, the uniform entangled state on the point basis, and controlled P_m access. Then:*

(a) $Q^m + 1 - 2gQ^{m/2} \le n_m \le Q^m + 1 + 2gQ^{m/2}$.

(b) Its Hadamard acceptance probability is $a = 1/2 + N_j/(2n_m)$, and the output is $\widehat{N}_j = 2n_m(\widetilde{a} - 1/2)$.

(c) $|\widehat{N}_j - N_j| \le \kappa$ exactly when $|\widetilde{a} - a| \le \kappa/(2n_m)$. With $0 < \kappa \le 2n_m$, choosing $M \ge 4\pi n_m/\kappa$ and the median as in Theorem 1 suffices, with $O(j n_m \kappa^{-1} \log(2/\eta))$ calls.

(d) For the minimally padded cohomological route $n = 2^{\lceil \log_2(2g) \rceil}$, the corresponding acceptance-probability tolerance for estimating N_j within κ is $\kappa/(2nQ^{j/2})$. The ratio of the cohomological tolerance to the permutation tolerance is exactly $n_m/(nQ^{j/2})$.

Without $j \mid m$, parts (b)–(c) target $N_{\gcd(j,m)}$ instead; a comparison for the same quantity must change the cohomological order accordingly. For $j = m$, $P_m^j = I$ and the test accepts with certainty.

Proof. Part (a) is the Weil bound. Apply the Bell-state identity to Proposition 3 to obtain (b). Subtraction of the true acceptance probability gives $\widehat{N}_j - N_j = 2n_m(\widetilde{a} - a)$, proving the exact equivalence in (c); the sufficient cost is the proof of Theorem 1 with n_m, κ substituted. In the cohomological route $\widehat{s}_j - s_j = 2nQ^{j/2}(\widetilde{a} - a)$, and estimating s_j or $N_j = Q^j + 1 - s_j$ has the same absolute error. Divide the tolerances to obtain (d). The remaining assertions follow from the general gcd formula. □

Rounding recovers an integer count whenever the achieved absolute error is strictly below $1/2$. For fixed g, Q and increasing m, $n_m \sim Q^m$; uniformly in g the dimension comparison requires $2gQ^{-m/2}$ small. At fixed confidence and fixed $\kappa < 1/2$, the coherent route's sufficient query count is $O(jn_m)$. Ordinary independent Hadamard sampling has sufficient cost $O(n_m^2)$ at this precision. Neither is a lower bound for arbitrary algorithms using P_m, and oracle queries cannot be equated to classical field-operation costs without an implementation model. The route also needs N_m and point-state preparation. Consequently "no better than classical counting" is not asserted as a theorem.

Validation v6. Cost comparisons now restrict to $j \mid m$. A regression at $j = 2, m = 3$ catches the former mismatch: on $y^2 = x^3 + x + 1$ over $\mathbb{F}_5$, $N_1 = 9$ and $N_2 = 27$, and that permutation test targets 9, not 27. The search for an amplitude-estimation register size uses a 361-point grid on $[0.05, 0.95]$. Its output is the smallest tested power-of-two size meeting that grid criterion, *not* a certified optimum or guarantee over the whole interval. The continuum upper bound rests on the theorem.

7 The cohomological circuit and classical simulation

Proposition 5 (Computing the polynomial and describing the oracle)**.** *Let $q = \lceil \log_2(2g) \rceil$, $n = 2^q$, and $\varepsilon_g = (2nQ^{g/2})^{-1}$.*

(a) Given L_C, approximate circuits for $\Lambda_1(V)$ and $\Lambda_2(V)$ for a diagonal realization can be produced classically in time polynomial in $g, \log Q, \log(1/\varepsilon)$. They need no private ancillas. Before finite-gate compilation, a diagonal synthesis of $\Lambda_1(V)$ uses at most $2^{q+1} - 1$ phase rotations and $(q-1)2^{q+1} + 2$ CNOTs, hence $O(g \log g)$ gates; the analogous construction for $\Lambda_2(V)$ uses one more control qubit.

(b) Suppose a classical description of O_1 of size S on $1 + q + a$ qubits satisfies Definition 2 with the promised *accuracy $\varepsilon \le \varepsilon_g/6$. Assume bounded-arity gates whose entries are computable to b bits in time polynomial in b and their description length. Then L_C can be recovered classically in time polynomial in $S, 2^{q+a}, g, \log Q$ and the gate-description length.*

(c) Under these encoding assumptions, constructing such accurate oracles with $a = O(\log g)$ in classical polynomial time is equivalent to computing L_C in classical polynomial time. The accuracy promise and effective gate descriptions are part of the equivalence.

(d) A quantum preprocessing algorithm can compute L_C using Kedlaya's algorithm [5], then apply (a). This is one construction route, not a necessity claim about all quantum constructions.

Proof. (a) Integer polynomial root isolation and refinement, keeping multiplicities, determine normalized roots to arbitrary precision in polynomial bit complexity. Project sufficiently accurate approximations to the unit circle. A diagonal unitary on $d = q + 1$ qubits with phases θ_x has the Walsh expansion

$$\theta_x = \sum_{S \subseteq \{1,\dots,d\}} c_S (-1)^{S \cdot x}, \qquad c_S = 2^{-d} \sum_x \theta_x (-1)^{S \cdot x}.$$

Its circuit is $e^{ic_\varnothing} \prod_{S \neq \varnothing} e^{ic_S Z_S}$. Compute each parity on one of its qubits with $|S| - 1$ CNOTs, apply a rotation, and uncompute. Summing gives $2^d - 1$ rotations and $\sum_{S \neq \varnothing} 2(|S| - 1) = (d-2)2^d + 2$ CNOTs. The same construction treats both supplied controls directly. Whole-circuit phases are allowed by Definition 2. Allocate error among root approximations and gate synthesis. Standard effective approximation over a fixed inverse-closed universal gate set adds a polynomial in $\log(g/\varepsilon)$ overhead [7].

(b) First consider exact arithmetic. Simulate the described circuit and form its clean-ancilla compression $B = (I \otimes \langle 0^a|) O_1 (I \otimes |0^a\rangle)$. This is a classical matrix operation, not a quantum

postselection assumption. Its control blocks obey

$$\|B_{00} - e^{i\gamma} I\| \le \varepsilon, \qquad \|B_{11} - e^{i\gamma} V\| \le \varepsilon.$$

Write $z = \mathrm{Tr}(B_{00})/n$. Then $|z - e^{i\gamma}| \le \varepsilon < 1$, so $z \neq 0$ and $\widehat{\phi} = z/|z|$ satisfies $|\widehat{\phi} - e^{i\gamma}| \le 2\varepsilon$. Let $V' = B_{11}/\widehat{\phi}$. It is a contraction and $\|V' - V\| \le 3\varepsilon \le \varepsilon_g/2$. Telescoping powers of contractions yields

$$|\,\mathrm{Tr}(V'^k) - \mathrm{Tr}(V^k)| \le nk\|V' - V\|.$$

Take *real parts* and remove padding:

$$\widehat{s}_k = Q^{k/2}\,\mathrm{Re}(\mathrm{Tr}(V'^k) - (n - 2g)).$$

For $k \le g$, its error is at most $nkQ^{k/2}\varepsilon_g/2 \le k/4$.

Now compute these values numerically with additional certified absolute error less than $k/8$. The total is less than $3k/8 < k/2$, so the congruence decoder recovers the exact integer coefficients. Such precision needs polynomially many bits: $O(g \log Q)$ bits cover the scaling, with additional polynomial terms for matrix dimensions, S, and gate descriptions. Standard error bounds for sums and products, and the bound $|z| \ge 1 - \varepsilon$, control the phase division. Full matrix simulation costs $O(SD^2)$ arithmetic operations for bounded-arity gates on $D = 2^{1+q+a}$ basis states; polynomial precision yields the stated bit bound. The fixed accuracy margin avoids dependence on an unspecified gap to a strict threshold.

(c) Apply (a) with $\varepsilon \le \varepsilon_g/6$ in one direction and (b), with 2^{q+a} polynomial in g, in the other. Part (d) follows from [5] and (a). □

Corollary 1 (Consequence of an explicit small circuit)**.** *Given an explicit circuit meeting Proposition 5(b) with polynomial gate-description length, polynomial size and $O(\log g)$ private ancillas, classical simulation recovers L_C in polynomial time. This conclusion concerns an explicit description; it does not apply automatically to opaque black-box access or to polynomially many ancillas.*

Conjecture 1 (General classical construction with logarithmic ancillas)**.** For all dense squarefree hyperelliptic inputs $y^2 = f(x)$ over odd-characteristic $\mathbb{F}_Q$, there is no uniform deterministic classical algorithm, polynomial in g and $\log Q$, that constructs an oracle of Definition 2 with n minimally padded, $a = O(\log g)$, polynomial description length, effective gates, and $\varepsilon \le \varepsilon_g/6$.

This is a proposed complexity conjecture, not a theorem and not a conclusion from the experiments. Under the hypotheses above it is equivalent to the corresponding deterministic classical polynomial-time nonexistence conjecture for computing L_C. The equivalence does not establish either nonexistence claim. Efficient special families do not contradict a worst-case assertion over all inputs. A randomized version would require specifying its success guarantees separately.

The blanket statement that no Frobenius-spectrum circuit can be built without first computing L_C is withdrawn. Van Dam [6] proposed a spectral quantum approach and constructed efficient zeta-spectrum circuits for special Fermat hypersurfaces. Those results must be acknowledged, while their work registers, invariant subspaces and control contracts require separate comparison with Definition 2. No claim is made here that they satisfy the logarithmic-private-ancilla promise in Proposition 5. General constructions with more workspace remain outside its classical simulation conclusion.

Validation v7. Gate-list synthesis, simulation of disguised circuits, and finite-precision recovery are tested on small instances. Robustness uses $\|O - \gamma\Lambda_1(V)\|$ for a chosen phase γ as a numerical *upper bound* on the best-phase distance. Trace alignment minimizes the Frobenius norm, not generally the operator norm. The new counterexample test records this distinction. Recovery is checked below the conservative $\varepsilon_g/6$ threshold. These machine-precision tests illustrate the theorem; they do not implement arbitrary-precision certification for all input sizes and do not test Conjecture 1.

8 Reproducibility and validation boundaries

The accompanying package retains seven result-specific scripts, their common arithmetic module, and a runner. The JSON results and logs are generated afresh, and the environment is recorded automatically. Exact finite-field enumeration and rational/integer identities are distinguished from numerical eigenvalues, matrix simulations and probability calculations. The latter use declared tolerances. Fixed seeds make experiments repeatable; floating-point results can differ slightly across library versions.

Script	Scope	Seconds	Result
v1	Intrinsic traces and point counts	18.1	PASS
v2	Unitary realization and exact Weil classes	32.4	PASS
v3	Estimator and approximate access	8.5	PASS
v4	Records and full-order curve witnesses	9.8	PASS
v5	Point permutation and fixed points	0.6	PASS
v6	Divisibility-correct route comparison	36.1	PASS
v7	Circuit synthesis and classical recovery	2.1	PASS

Table 2: Corrected reference run. Timings depend on hardware.

Reference environment: CPython 3.12.14, NumPy 2.3.5, SciPy 1.17.0, SymPy 1.14.0, and mpmath 1.3.0.

Run the package in a virtual environment, so the runner's isolated Python mode can find its dependencies:

```
python3 -m venv .venv
.venv/bin/python -m pip install -r requirements.txt
.venv/bin/python -I run_all.py
```

The Windows equivalents use `.venv\Scripts\python.exe`. The environment file and pinned requirements describe this corrected reference run, not the earlier supplied environment. Verify the packaged files with `sha256sum -c SHA256SUMS` before running, since a run updates result files and timing metadata.

The scripts validate examples and implementation consistency. They neither certify all possible curve inputs nor replace the mathematical proofs. They establish no algorithmic lower bound, no universal realizability result for Weil polynomials, no continuum optimum from a finite grid, and no resolution of Conjecture 1. The exact number of tested cases, runtime and tolerances are in the supplied JSON and logs, rather than inferred from a prior run.

8.1 Exact certification of the finite Weil census

The revised `is_weil_exact` routine uses integers only. Given the functional equation, write

$$T^{-g}L(T) = h(T^{-1} + QT), \quad h(y) = b_g + \sum_{k=0}^{g-1} b_k V_{g-k}(y),$$

where $V_0 = 2$, $V_1 = y$, $V_d = yV_{d-1} - QV_{d-2}$. The monic integer polynomial h has roots y_i. Its Newton sums $p_k = \sum_i y_i^k$, including $p_0 = g$, are exact integers. Form

$$H = (p_{i+j})_{0\le i,j<g}, \qquad K = (4Qp_{i+j} - p_{i+j+2})_{0\le i,j<g}.$$

Then $L \in W_g(Q)$ if and only if both matrices are positive semidefinite, after checking integrality, degree, constant coefficient and the functional equation.

Proof. For a real polynomial u of degree below g, its coefficient vector satisfies $u^T H u = \sum_i u(y_i)^2$. If all roots are real, this is nonnegative. If there is a nonreal conjugate pair, real polynomial interpolation at the distinct roots can prescribe values $i, -i$ on that pair and zero elsewhere. The resulting quadratic form is strictly negative, counting their positive multiplicities. Thus $H \succeq 0$ is equivalent to reality of every root, including repeated roots.

Once all roots are real, $u^T K u = \sum_i (4Q - y_i^2) u(y_i)^2$. This is nonnegative exactly when every $y_i^2 \le 4Q$: necessity follows by interpolating one at a chosen distinct root and zero at the others. Finally the roots of $x^2 - y_i x + Q$ have modulus $\sqrt{Q}$ exactly when $y_i \in [-2\sqrt{Q}, 2\sqrt{Q}]$, proving the criterion.

A real symmetric matrix is positive semidefinite exactly when every principal minor is nonnegative. The implementation evaluates these determinants by exact integer elimination. The coefficient bounds $|b_k| \le \binom{2g}{k} Q^{k/2}$ have exact integer floors $\mathrm{isqrt}(\binom{2g}{k}^2 Q^k)$. Enumerating those ranges and testing every candidate is complete; no numerical root filter can silently discard a candidate. This finite exhaustive routine is not claimed to be efficient as g, Q grow. □

9 Corrections in this edition

The following changes are incorporated in the statements, proofs and matching code:

1. Curve witnesses are selected and checked through order $2g$. The earlier six examples remain as a regression detecting four full-order failures. Abstract witness selection explicitly chooses the endpoint attaining (5).
2. Proposition 4 and same-target cost comparisons require $j \mid m$, while Proposition 3 retains the general gcd identity. Point-state preparation and coordinate-encoding costs are explicit assumptions.
3. Actual small-circuit oracle instrumentation is distinguished from formula-based bookkeeping. Approximate-access tests include separately perturbed controls, global phases and ancilla leakage. Finite amplitude-grid results are labeled as grid results.
4. Classical simulation uses real parts, effective gate descriptions, a fixed $\varepsilon_g/6$ promise and a separate numerical error budget. The noise test measures a phase-dependent upper bound, not an incorrectly claimed exact minimization.
5. Weil-class membership and coefficient bounds are now entirely exact. Numerical spectral routine names explicitly say `numeric`.
6. Cost bounds retain n unless minimal padding is assumed; the exact genus factor in the precision threshold is retained. Prior spectral work is cited and the universal "computes L first" claim is removed.
7. The exact requested attribution appears in every page header and footer and at the start and end of every Python file. The full MIT license is included in the paper, package and Python files.

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